Enrolment No.: 23104W012007

Institute of Infrastructure Technology Research and Management,
Ahmedabad - 380026, Gujarat

B.Tech. Semester VI - Class Test I

Date: 11th February 2026 **Total Marks:15**
Course Code: CS 243008 **Mobile Computing**

(1) Which of the following is not a component of Mobile Computing? [1]

(a) Smartphone (b) Laptop (c) Desktop (d) Wearables

(2) Which of these is not a terminology associated with Mobile IP? [1]

(a) Home agent (b) Secondary agent (c) Correspondent (d) Care-of-address

(3) Which of the following is not a mitigation technique for High bit-error rates? [1]

(a) Error correction
(b) increasing transmit power
(c) link-layer retransmission
(d) Ethernet bridging

(4) Which of the following is not a Random Access Protocol? [1]

(a) TDMA (b) Aloha (c) Slotted Aloha (d) CSMA

(5) The ISM band stands for ..... [1]

(6) The maximum throughput of ALOHA and Slotted ALOHA is ....0.186.....% and ..0.356.....% respectively. [1]

(7) Which of the following is likely a correct combination of SIFS and DIFS for CSMA/CA with Ack ? [1]

(a) DIFS: $10\mu s$, SIFS: $20\mu s$
(b) DIFS: $20\mu s$, SIFS: $10\mu s$
(c) DIFS: $50\mu s$, SIFS: $10\mu s$
(d) DIFS: $10\mu s$, SIFS: $50\mu s$

(8) In the context of CSMA/CA, RTS and CTS stands for .....Routing transmit system Computing transmit system..... [1]

(9) An amplifier is known to have a power gain of 200. Its power gain in decibels is ...23 dBm. [1]

(10) An amplifier is known to increase signal power from -51 dBm to -35 dBm. What is the gain/loss of the amplifier in dB? ...-16 loss. [1]

(11) For the above question, what is the gain/loss factor with which the input signal is multiplied? ..~~8~~..10..... [1]

(12) In CSMA without random backoff, suppose two transmitting stations collide. After collision, they always choose the same number of time slots to backoff and transmit again. What is the effective throughput of both the stations? . ~~CSMA/CA~~ . 0 [1]

(13) Out of Non-persistent CSMA, 1-persistent CSMA and p-persistent CSMA, collision probability is highest in ....1-persistent CSMA . [0.5]

(14) Out of Non-persistent CSMA, 1-persistent CSMA and p-persistent CSMA, channel utilization is highest in ....Non-persistent CSMA - [0.5]

(15) 0.1-persistent CSMA has higher throughput than 0.5-persistent CSMA. True or False. Justify. [2]

True, because 0.1-persistent CSMA is greater then 0.5 persistent CSMA-